Detecting decline before diagnosis

Wearables, digital biomarkers and neurological research

Evidence review | 20 October 2026

Executive findings

Consumer devices can measure clinically relevant changes in movement, cognition and speech. Some signals precede a recorded diagnosis. But measurement, prognosis, disease diagnosis and improved patient outcomes are four different claims; few systems establish all four.

• **Parkinson's:** wrist accelerometry identified a risk signal up to seven years before diagnosis. This was retrospective prediction within a prospective cohort, not a deployed screening trial. Its precision–recall performance remains insufficient for autonomous diagnosis.

• **Alzheimer's:** remote cognition has direct associations with amyloid/tau PET. A newer primary-care study independently validated digital cognition combined with blood biomarkers. This is more clinically actionable than smartwatch-only claims.

• **Monitoring:** the strongest evidence concerns repeated assessment of known disease, treatment fluctuations and trial endpoints—not screening everyone for any neurological disease.

• **EEG:** sleep physiology relates to PET/CSF pathology and subsequent decline, but this does not validate every consumer headband or proprietary sleep score.

• **Usability:** charging, onboarding, reminders, accessibility and caregiver support determine whether an otherwise valid endpoint can be collected.

• **Infrastructure:** Qualtrics plus REDCap is a feasible research design, not an established universal standard. Security depends on the actual deployment, contracts and workflow.

Best-supported direction: a disease-specific, longitudinal triage or monitoring service linked to clinical confirmation. The evidence does not establish a general-purpose device that diagnoses "bodily decline."

Research synthesis, not individual medical advice or a regulatory determination.

1. Scope and evidence rules

This is a targeted scoping review, **not an exhaustive systematic census of every trial**. The principal window is October 2016–October 2026; earlier studies continuing into that window and foundational infrastructure papers are identified where relevant. Neurology is the focus, with selected neuromuscular examples—not every possible early diagnosis across medicine.

Sources are restricted to the requested families: PubMed/NLM, ClinicalTrials.gov, Elsevier/ScienceDirect, Springer Nature, Wiley, IEEE Xplore, JMIR, Qualtrics and REDCap. Articles from other publishers are linked only through permitted PubMed/NLM records. No company promotional evidence is used to establish diagnostic performance.

Searches combined disease terms (Parkinson's, Alzheimer's, MCI, MS, Huntington's, ALS, ataxia) with wearable, accelerometer, smartphone, gait, speech, EEG, digital cognition, PET, CSF, EMA and adherence. PubMed metadata and abstracts were retrieved through NLM E-utilities; trial designs, dates, outcomes and status were checked through the ClinicalTrials.gov API. Official product documentation supports the

infrastructure section. Full-text methods were not independently audited for every paper. Metrics below are reported study estimates, not pooled estimates.

Evidence categories

1. **Measurement validation:** the signal agrees with a reference instrument or clinical construct.

2. **Cross-sectional discrimination:** existing cases differ from controls.

3. **Longitudinal prediction:** baseline signals predict later outcomes.

4. **Clinical utility:** acting on the signal improves decisions or outcomes.

A study can succeed at level 1 or 2 without establishing level 3 or 4. A biomarker can be associated with disease without being a validated surrogate for treatment benefit. Active performance tests may be digital clinical outcome assessments rather than biological biomarkers in the strict sense.

2. What changed during the decade

2016–2019: feasibility and case–control separation. Smartphone tapping, gait and voice studies showed measurable differences and medication sensitivity. Early home-dementia studies were small and heterogeneous. A 2019 systematic review found only 26 eligible home-based studies, ranging from 12–279 participants and three days–3.6 years [24].

2020–2022: validation frameworks and remote cohorts. V3 formalized verification, analytical validation and clinical validation [25]. Floodlight linked smartphone performance to MS clinical and MRI measures. Huntington's batteries demonstrated reliability across active motor and cognitive tasks. AT-HOME PD connected virtual follow-up to participants from completed drug trials [22].

2023–2024: longer follow-up and biomarker anchoring. UK Biobank demonstrated a prediagnostic Parkinson's signal. Mayo remote cognition studies used amyloid/tau PET reference groups. WATCH-PD measured 12-month change rather than merely distinguishing cases from controls. Self-supervised accelerometer methods improved gait recognition.

2025–2026: multimodal clinical workflows and usability validation. BioCog combined digital cognition with blood testing in an independent primary-care cohort. RADAR-AD reported modest rather than near-perfect prodromal discrimination. V3+ added usability validation. Current registries increasingly pair digital data with imaging and fluid biomarkers, but many have no posted results.

The literature is expanding faster than the evidence for clinical utility. A 2025 early-Parkinson's review found 12 eligible studies from 1,044 deduplicated records; ten were cross-sectional [23]. Another review found 15 longitudinal progression studies, including four interventional studies; rating scales remained the usual comparator [5].

Main lesson: research is progressing from "can a device separate groups?" toward "can a validated change measure support a decision in an individual?"

3. Parkinson's: prediction versus monitoring

Seven years before diagnosis—what that actually means

The 2023 Nature Medicine study used UK Biobank wrist accelerometry to distinguish diagnosed and subsequently diagnosed Parkinson's from a general-population comparison group [1]. The reported comparison

included 153 diagnosed cases, 113 prodromal cases and 33,009 population controls. Prediagnostic measurements extended up to seven years before recorded diagnosis.

For prodromal disease, the area under the precision–recall curve (AUPRC) was **0.07 ± 0.03**, versus approximately 0.01 for genetics, 0.03 for lifestyle, 0.01 for blood biochemistry and 0.01 for recorded prodromal signs. The diagnosed-case AUPRC was 0.14 ± 0.04.

This is not 93% diagnostic accuracy, nor 7% positive predictive value at a fixed threshold. AUPRC summarizes multiple thresholds. The relevant finding is relative enrichment in a rare condition. The study does not show that every future patient was detectable seven years early, or that the system improves outcomes when used prospectively.

Following established disease

mPower smartphone tasks—especially tapping—predicted self-reported Parkinson's status with ROC AUC 0.80 and correlated with clinical motor severity at r=0.71 [2]. Self-reported labels and motivated smartphone volunteers limit diagnostic claims.

An Oxford study followed 237 people with early Parkinson's. Seven-minute smartphone tests predicted falls, freezing, postural instability, cognitive and functional impairment at 18 months [3]. AUC exceeded 0.90 in reported 10-fold analyses using all features; reduced 30-feature models exceeded 0.75. Internal validation is not independent clinical deployment.

These studies support repeated motor assessment and risk stratification within known Parkinson's—not a universal micro-tremor diagnostic.

What WATCH-PD adds

WATCH-PD enrolled 82 people with early untreated Parkinson's and 50 controls. At home, participants wore a smartwatch for seven days after clinic visits and completed smartphone motor, speech and cognitive tasks every other week [4].

Over 12 months, average in-clinic arm swing decreased from 25.9° to 19.9°, while the proportion of awake time with tremor increased from 19.3% to 25.6%. Changes generally exceeded corresponding individual MDS-UPDRS items, **not the overall scale**. Some analyses were sharply reduced by medication initiation and missing data; the reported daily-step analysis had only ten participants. This is evidence of responsiveness, with practical limitations—not proof that all digital endpoints outperform clinical examination.

Tremor, gait and speech are not interchangeable

Small-amplitude tremor occurs in healthy physiology and in essential tremor, medication effects and other conditions. Parkinson's may occur without prominent tremor. The context, frequency, amplitude, rest/action state and associated bradykinesia matter.

A 32-person smartphone study used accelerometer/gyroscope signals transformed into frequency-domain images for a CNN. It reported 93.8% sensitivity and 90.1% specificity for freezing episodes [6]. This detects an event in people with Parkinson's; it does not diagnose preclinical disease.

A web-based speech study of 726 participants achieved AUC 0.753 for self-reported Parkinson's using acoustic features and XGBoost [7]. Its practical value is accessibility, not disease specificity. Microphone, language, noise, respiratory disease and medication can influence the result.

The Lancet Neurology review on unsupervised mobility explains why clinic performance and real-world mobility differ: psychological, physiological, cognitive, environmental and technical factors all contribute [21]. Home data

add information; they are not simply a noisier copy of the clinic test.

4. Alzheimer’s: links to PET, CSF and blood biomarkers

Remote cognition versus amyloid/tau PET

Mayo Test Drive was evaluated in 684 adults, 94% cognitively unimpaired [8]. Lower remote performance was associated with greater amyloid and tau PET burden, smaller hippocampal volume and more white-matter hyperintensity. The remote test measures cognition; PET anchors the interpretation to pathology. The design was cross-sectional, so the association alone does not prove future decline prediction.

In a related 353-person study, remotely administered Stricker Learning Span performed similarly to the in-person Auditory Verbal Learning Test in separating amyloid-positive and amyloid/tau-positive groups [9]. PET could be up to three years from testing: temporal mismatch is an important limitation. Related Mayo studies should not be treated as wholly independent replications without examining cohort overlap.

A smaller study of 69 cognitively normal adults used smartphone tasks three times daily over eight days [10]. The episodic-memory task achieved AUC 0.77 for amyloid status, compared with 0.74 for MoCA. That numerical difference is not evidence of universal superiority.

Learning rate can be an endpoint

In 114 cognitively unimpaired adults, monthly at-home testing showed expected practice-related improvement [11]. Smaller improvement over the first three months correlated with greater amyloid burden (r=-0.20) and entorhinal tau (r=-0.38). The early learning slope discriminated subsequent PACC-5 decliners with AUC 0.91; the one-year in-person follow-up subset contained 72 participants.

This suggests a useful mechanism: early disease may impair the capacity to benefit from repeated exposure before a single test becomes abnormal. But a small cohort and a specific definition of decline require replication. Repeated testing must model learning explicitly rather than interpreting every score increase as recovery.

Digital cognition plus blood: closer to a usable pathway

BioCog, published in Nature Medicine in 2025, developed a model in 223 secondary-care participants and evaluated it in 403 participants across 19 primary-care centers [12]. Digital cognition identified cognitive impairment with 85% accuracy using one cutoff, versus 73% for primary-care physicians. BioCog plus blood testing identified clinical, biomarker-verified AD with 90% accuracy, compared with 70% for standard care and 80% for blood alone.

This independent-cohort design is stronger than random splits of a single dataset. It remains a proof-of-concept in a clinical population, not asymptomatic population screening. A two-cutoff approach reported higher cognitive-classification accuracy, but deployment must also report how many people fall in the intermediate zone and require further evaluation.

Speech prediction

A Framingham study analyzed neuropsychological interview speech from 166 people with MCI: 90 progressed and 76 remained stable [13]. Language-model features plus demographics predicted progression within six years with 78.5% accuracy and 81.1% sensitivity. This is prognosis in an already impaired population. It is not six-year prediction from ordinary conversation in healthy adults, and the main model used recognized text rather than exclusively raw vocal acoustics.

Consumer devices at scale

The Intuition study publication describes 23,004 adults using iPhones and Apple Watches over up to 24 months [14]. Its initial results concern interactive cognitive assessment and MCI classification, not proof that passive watch data predict AD. The registry reports 22,720 enrolled and a terminated study; publication and registry counts differ. They should not be silently reconciled.

RADAR-AD combined multiple remote technologies. Its 2025 analysis reported prodromal discrimination at AUC 0.73 [15]. Movement, sleep, gait and cognition contributed, but adding modalities did not produce near-perfect diagnosis.

5. EEG, MS and other neurological conditions

EEG: biology is promising; consumer translation is incomplete

A sleep study linked impaired slow-oscillation/spindle coupling to medial temporal tau PET and diminished low-frequency slow-wave activity to amyloid PET [16]. These were polysomnographic measures—not an ordinary smartwatch sleep score.

A 2025 study combined sleep EEG, CSF AD biomarkers, MRI and two-year cognition in 93 participants [17]. Disrupted coupling across slow oscillations, theta bursts and spindles related to pathology and predicted accelerated cognitive decline. This is a direct example of physiological signals alongside CSF, but not a validated consumer-headband screening product.

A commercial 16-channel EEG study distinguished mild AD, non-AD MCI and controls using automated artifact rejection, power/entropy/complexity features and a multilayer perceptron. It reported F1=0.88 under leave-one-subject-out validation [18]. The study was preliminary and cross-sectional. A 16-channel system is not equivalent to a two- or four-channel consumer headband.

Multiple sclerosis

Floodlight assessed cognition, pinching, drawing, gait and balance over 24 weeks [19]. Published results included 76 people with MS and 25 controls; the registry reports 100 enrolled. Test–retest ICCs were 0.61–0.85. Digital cognition correlated with its clinical comparator at r=0.82; several measures related to disability and brain volume. MRI, rather than amyloid PET, is the disease-relevant imaging anchor.

A 68-person MS study combined accelerometer, gyroscope and surface EMG during finger/foot tapping [20]. Derived metrics separated relapsing from secondary-progressive MS with AUROCs 0.93 and 0.96. Small-group separation and change despite stable EDSS are promising, but do not establish reliable individual diagnosis of progression.

Huntington’s, ALS/FTD and neuromuscular disease

A Huntington’s platform combined six motor and two cognitive tasks. Across three studies, 20 controls, 20 premanifest and 179 manifest participants contributed data [27]. Sensor features had ICCs of 0.89–0.98, and digital cognitive tests correlated with in-clinic equivalents at r=0.79–0.90. High reliability is useful for trials but does not prove a drug benefit or identify an optimal treatment threshold.

PROSA is a prospective ALS/FTD speech-biomarker protocol targeting 75 ALS patients, 75 FTD patients and 50 controls over 12 months [28]. Its automated telephone tasks address motor, respiratory and cognitive function. The cited paper is a protocol: expected diagnostic/prognostic performance is not a published result.

Nature Medicine studies used full-body motion capture in Friedreich's ataxia and Duchenne muscular dystrophy [29,30]. The ataxia study included only nine patients and nine controls; DMD included 21 patients and 17 controls. They suggest richer motion descriptors can predict progression more precisely than selected conventional scores. They are hypothesis-generating, research-grade systems, not consumer-device validation or early-diagnosis trials.

Across diseases: the best reference standard changes. AD requires pathology-aware cognitive interpretation; PD studies often use clinical examination, dopamine-transporter imaging and CSF markers; MS uses disability, MRI and fluid measures; Huntington's has genetic confirmation, clinical scales and CSF mutant huntingtin. "PET/CSF" should not be treated as one interchangeable gold standard.

6. Observational versus interventional evidence

Having digital measurements inside a drug trial does not establish them as validated surrogate endpoints. Demonstrating reduced tremor after medication is different from showing slowed neurodegeneration. A measurement-validation study may also be registered as interventional because it assigns devices, despite testing no therapeutic effect.

A direct CSF-linked example—and a warning

The Huntington's digital-platform validation used predrug screening data from the tominersen extension NCT03342053 and participants from the HD Natural History Study NCT03664804 [27]. The extension measured CSF drug concentration, mutant huntingtin, MRI and clinical outcomes. This links digital collection to a biomarker-rich therapeutic program, but the validation paper's predrug analysis does not demonstrate treatment responsiveness.

The natural-history registry contains a particularly important limitation: new stability information rendered its CSF mutant huntingtin samples unsuitable for certain endpoints, so no valid data were available for those analyses. Other clinical, MRI and neuronal-injury outcomes were posted. **A technically successful sensor cannot rescue an invalid reference assay.** Sample stability, assay batches and processing delays belong in the data model.

Device interventions

A randomized pilot of 40-Hz light/sound stimulation in mild probable AD enrolled 15 participants and assessed EEG entrainment, compliance, MRI, cognition and activity rhythmicity [31]. Exploratory findings favored stimulation on selected measures. It was small, not definitive efficacy evidence, and did not establish human amyloid/tau lowering merely because animal studies had done so.

A soft robotic garment trial in one person with Parkinson's repeatedly reduced freezing over six months [32]. It provides proof of mechanism for assistive technology, not generalizable treatment efficacy or early detection.

What is still missing

The reviewed evidence does not establish a replicated, large trial in which consumer-sensor screening of undiagnosed people, followed by a prespecified clinical response, improves neurological outcomes. Such a trial must measure referral burden, false positives, time to diagnosis, treatment eligibility, anxiety, costs and patient benefit—not only algorithm accuracy.

7. Verified registry map

Status is the registry's label when checked on 20 October 2026, not confirmation that a site is currently enrolling. Estimated enrollment is marked **planned**. "No posted results" does not mean no associated publication. Conversely, completion does not establish success. These 16 records are a high-relevance selection, not all worldwide trials.

Current cohorts and validation programs

Study and period	Design / enrollment	Measurements and relevance	Registry status / evidence
PPMI, NCT04477785, 2020–2033 estimated	Observational; 4,500 planned	Digital, clinical, imaging, biological and genetic progression data; manifest and prodromal PD	Recruiting; updated July 2026; no posted results for this record
ADNI4, NCT05617014, 2023–2027 estimated	Observational; 1,500 planned	Remote and clinic cohorts; cognition, amyloid/tau PET, MRI, blood and CSF	Enrolling by invitation; updated September 2026; no posted results
Bio-Hermes-002, NCT06584357, 2024–2028 estimated	Cross-sectional; 1,200 planned	Explicit digital, blood, CSF, retinal, MRI and amyloid/tau PET database	Recruiting; updated April 2026; no posted results. Biomarker association, not years-long progression
Digital cognition + blood triage, NCT07041450, 2025–2026 estimated	Prospective observational; 500 planned	Time to determining anti-amyloid eligibility; digital cognition and blood, modeled follow-up testing costs	Recruiting label; last update July 2025; estimated completion already passed. Verify with sponsor

Study and period	Design / enrollment	Measurements and relevance	Registry status / evidence
DreaMS validation, NCT05009160, 2022–2027 estimated	Observational; 300 planned	App challenges, monitoring and surveys compared with clinical, imaging and body-fluid assessments	Active, not recruiting; updated March 2026; no posted results
Smart-insole MS monitoring, NCT05781113, 2023–2026 estimated	Observational; 90 planned	Three 15-minute walking samples/week; adherence and clinician acceptance primary; gait change secondary	Active, not recruiting; updated March 2026; no posted results
MARKERS-NDD, NCT06596746, 2024–2034 estimated	Observational; 600 planned	Movement analysis, imaging, EEG, cognition and blood chemistry across neurodegenerative conditions	Recruiting; updated September 2026; no posted results
Passive sensing/NR substudy, NCT05245903, 2022–2026 estimated	Observational substudy; 40 planned	Ambient Emerald sensing of sleep, gait and diurnal activity during parent nicotinamide-riboside study	Recruiting label; updated November 2025; estimated completion passed. Not a consumer wearable or proven PET/CSF validation

Study and period	Design / enrollment	Measurements and relevance	Registry status / evidence
PASADENA, NCT03100149, 2017–2031 estimated	Phase 2 prasinezumab; 316 actual	Clinical MDS-UPDRS and DaT-SPECT; biomarker-rich intervention program	Active, not recruiting; results posted. DaT-SPECT is not PET; registry status includes extensions, not only the initial controlled period

Completed or terminated studies

Study and period	Design / enrollment	Measurements	Verified interpretation
Real-PD, NCT02474329, 2015–2016	Observational; 304 actual	Smartwatch/pendant accelerometry; medication and mood; clinical assessment	Completed; results posted; 291 contributed 13-week data. Started before review window but foundational
Floodlight PoC, NCT02952911, 2016–2018	Nonrandomized device feasibility; 100 actual in registry	Smartphone/watch active tests and passive monitoring	Completed; no tabular registry results, but published validation [19]. Registered interventional does not imply a therapeutic trial
Intuition, NCT05058950, 2021–2023	Observational; 22,720 registry enrollment	iPhone/Apple Watch; interactive cognition and passive-use objectives	Terminated; no posted registry results; publication [14] reports 23,004 and initial interactive-task findings
Altoida, NCT06223438, 2024–2025	Observational; 614 actual	Multimodal sensor tasks, MCI classification and amyloid-related aims	Completed; no posted results. AUC 0.75–0.80 is a registered target, not achieved performance

Study and period	Design / enrollment	Measurements	Verified interpretation
FUTURE extension, NCT04846426, 2020–2024	Prospective observational; 67 actual	Speech prediction of PACC5 change; clinically and amyloid-stratified groups	Terminated; no posted results. Termination alone is not evidence that speech prediction failed
Tominersen extension, NCT03342053, 2018–2019	Randomized open-label phase 2 extension; 46 actual	CSF drug/mutant huntingtin, MRI, EEG, cognition; predrug digital subset [27]	Completed; results posted. Digital validation and treatment benefit remain separate claims
HD Natural History, NCT03664804, 2018–2022	Longitudinal cohort; registry labels interventional phase 1; 95 actual	CSF mutant huntingtin/NfL/tau, MRI and clinical scales; digital subset [27]	Completed; results posted; certain mutant-huntingtin analyses invalidated by sample stability

How to interpret this map

• The most direct current digital–PET/CSF match is **Bio-Hermes-002**, but it is primarily cross-sectional.

• **ADNI4** supplies longitudinal cognitive, imaging and fluid infrastructure; not every remotely enrolled participant undergoes every biomarker procedure.

• **PPMI** is a strong disease-specific discovery framework; confirm exact digital and biomarker availability for each subcohort.

• **DreaMS** asks whether digital changes correspond to clinical worsening, not merely whether patients differ from controls.

• **Huntington's** offers co-collected digital and CSF data, but assay limitations and predrug analysis constrain conclusions.

• Many relevant digital endpoints are exploratory or appear in substudy publications rather than the registry's primary outcome list. An exhaustive review requires protocol and publication linkage, not keyword matches alone.

8. From sensor to clinical endpoint

The following is an engineering synthesis, not one validated pipeline. Every parameter must be justified for the device, symptom and intended use [6,25,26,33].

Sensor design

Modality	Useful measurements	Main design trade-off
Wrist accelerometer/gyroscope	Tremor, arm swing, activity, turning	Comfortable, but arm motion confounds gait; raw access and sampling may differ by device
Foot/shank/waist IMU; pressure insole	Gait timing, asymmetry, freezing, balance	Better movement localization, but shoe fit, positioning and multi-device burden matter
Phone touchscreen/IMU	Tapping, dexterity, gait, reaction time	Broad access; handset size, operating system, placement and dexterity affect results
Microphone	Phonation, articulation, pauses, language	Low cost; privacy, language, noise and microphone changes create domain shift
EEG	Oscillations, event responses, sleep rhythms	Electrode contact, montage, artifacts and overnight comfort constrain validity
Optical pulse sensor	Heart rate, selected sleep/autonomic proxies	Motion, perfusion and skin-related performance differences; not direct brain measurement

Acquisition and quality control

Retain raw data where consent, storage and hardware permit. Record device model, placement, firmware, app and algorithm version, effective sampling rate, units, timestamps and medication context. Synchronize clocks; detect dropped packets, clipping, nonwear and low-quality segments before inference.

Illustrative design ranges—not universal standards—are 25–100 Hz for many activity/gait tasks, often 50–100 Hz or more for tremor characterization, 16 kHz or higher for speech, and 128–256 Hz or higher for EEG depending on the endpoint. Nyquist is only a minimum: noise, anti-aliasing and transient events also determine requirements. Sampling more aggressively trades battery life and adherence against information.

Signal processing and models

Movement: calibrate axes, handle gravity and orientation, identify wear/context, then segment into tasks or walking bouts. Derive cadence, stride timing variability, turning velocity, arm swing, spectral power, jerk and entropy. Short windows can recognize events; longer aggregation is needed for stable daily endpoints. Avoid filtering away low-frequency movement or labeling vehicle vibration as tremor.

Raw-signal deep learning: 1D CNNs/ResNets learn local temporal patterns directly from multiaxis windows; temporal convolutions, recurrent networks or transformers can add sequence context. Self-supervised pretraining uses unlabeled recordings before fine-tuning on smaller labeled cohorts. Models still need calibration, quality control and clinically valid labels.

ElderNet provides a relevant example [33]: self-supervised wrist accelerometry was evaluated using more than 1,000 unlabeled participants and 83 labeled participants across clinical groups. Reported gait-detection accuracy was 96.43%, but recall was 82.32% and F1 82.92%. **This is walking detection, not 96% accuracy for neurological diagnosis.**

Spectrogram models: the freezing study transformed inertial signals into frequency-domain images before CNN analysis [6]. It is not an end-to-end raw accelerometer model. This distinction matters when reproducing a pipeline.

Speech: retain quality metrics; distinguish acoustic analysis from automated transcription and language analysis. MFCCs, jitter, shimmer and pause features can feed gradient boosting; audio encoders or language models supply learned representations. Transcription errors can correlate with impairment, accent or microphone quality and become spurious predictors.

EEG: reject poor contact, line noise, eye and muscle artifacts; preserve valid rhythms before extracting power, coupling, connectivity or complexity. A classifier should not learn the recording site or sleep-apnea burden instead of AD pathology.

Validation that survives deployment

1. **Separate by participant before windowing.** Overlapping windows from one person must not appear in both training and test sets. Fit normalization and feature selection inside training folds.

2. **Validate by site, device and time.** A held-out random sample from one dataset is weaker than external prospective validation. Split families where genetic similarity could leak information.

3. **Compare simple baselines.** Age, education, baseline clinical status, medication and basic activity may explain much of the apparent AI gain.

4. **Measure calibration and clinical workload.** Report sensitivity, specificity, PPV/NPV, ROC AUC, precision–recall, uncertainty, referral rate and indeterminate results. Accuracy alone can obscure class

imbalance.

5. **Use longitudinal statistics.** Mixed-effects models can separate baseline differences from within-person slopes; survival models address conversion with censoring. Test practice effects, medication initiation, competing events and informative dropout.

6. **Anchor meaningful change.** A statistically detectable difference may be too small to matter. Compare with reliable-change limits, patient-reported function and clinically relevant progression.

7. **Lock the model and audit updates.** Firmware or app changes can create apparent decline. Freeze confirmatory versions and revalidate material changes.

Illustrative base-rate problem: with disease prevalence 1%, sensitivity 90% and specificity 90%, screening 10,000 people produces about 90 true positives and 990 false positives: PPV is only 8.3%. These are hypothetical numbers, not a study result. This is why enriched populations and confirmatory testing matter.

V3 separates hardware verification, analytical validation and clinical validation [25]. V3+ adds usability validation [26]. None alone establishes clinical utility or regulatory authorization for an intended diagnostic claim.

9. Adherence, usability and ecological assessment

A one-year Parkinson's remote-monitoring cohort enrolled 200 people with a fall history [34]. Only 116 (58%) completed the year. Median watch wear was 17.5 hours/day; technical problems accounted for 61 of 118 reported dropout reasons. Completion, wear time and analyzable sensor time are different denominators.

Floodlight Open provides unusually concrete UX evidence [35]. Simplifying activation increased the activation rate from 53.9% to 74.6%. Configurable reminders improved consecutive use and completion. These were study comparisons, not proof that one interface intervention will have the same causal effect in every population.

In Huntington's, active-task adherence across three cohorts was approximately 90%, 72% and 69% during the assessed short period [27]. Short, supported feasibility cannot be extrapolated to a year of unsupervised use. WATCH-PD participants generally found its technologies acceptable, yet the longitudinal study still experienced significant data loss [4].

EMA is not the same as cognitive testing

• **EMA:** contemporaneous self-report of fatigue, mood, pain, sleep quality, medication state or perceived cognitive difficulty.

• **Ecological momentary cognitive testing:** a brief performance task with measured accuracy, reaction time or learning.

• **Passive sensing:** observations collected without active task completion.

Combining them helps distinguish disease from context. A slower cognitive task after poor sleep or during an "off" medication period has a different interpretation from a persistent unexplained change.

Measure delivery, response latency, completion, valid wear, valid-task quality, support contacts and reasons for discontinuation separately. Missing data may reflect deterioration, technical failure or everyday life; do not automatically convert nonresponse into a disease signal.

Practical protocol choices

These are recommendations to pilot, not universally validated schedules:

• Use short measurement bursts when intensive cognition is needed, rather than assuming daily testing for years is sustainable.

• Offer accessible instructions, large targets, audio alternatives and caregiver assistance where valid; record assistance because it changes task interpretation.

• Supply devices/data plans where possible. Bring-your-own-device designs can exclude people with fewer resources and increase hardware variation.

• Let participants select reminder windows. Record postponements, late responses and time zones; avoid unrestricted retrospective completion of “momentary” surveys.

• Model practice, fatigue and ceiling/floor effects. Alternate forms need psychometric equivalence.

• Prespecify missing-data analyses, including sensitivity analyses for informative dropout, and do not impute absent symptoms as normal.

10. Qualtrics and REDCap: supported capabilities versus assumptions

Finding: the reviewed sources support both platforms as research infrastructure, but do not establish their combination as a standard neurological architecture for daily cognitive testing.

The Qualtrics mobile-EMA methods paper describes a hypothetical smoking study, not a neurological outcome trial [38]. A recent Par-Cog development paper used Qualtrics for expert feedback; its cognitive testing was a separate process. Neither validates the claimed daily-cognition/REDCap workflow.

REDCap can itself administer surveys and manage longitudinal data. Adding Qualtrics should solve a defined scheduling, interface or institutional requirement; otherwise it creates another transfer, identity mapping and failure point.

Important psychometric limit: putting a validated paper test into a survey does not validate its digital version. Browser timing, screen size, motor impairment, supervision and device latency can change the construct. For precise reaction-time or audio tasks, use a validated task engine and bring derived scores into the registry rather than assuming generic survey timing is sufficient.

Proposed architecture

REDCap: study system of record. Maintain consent, identity/contact information, eligibility, visit schedule, diagnoses, medications, adverse events, clinical scores and biomarker results. Use role-based permissions, controlled exports, data queries and audit trails [36,37,39]. Separate identifying and analysis-facing data as appropriate.

Qualtrics: participant-facing EMA, if justified. Use institutional licensing and approved configurations for brief surveys, context and usability feedback. Official personal links are unique and single-use, tied to recipients and potentially embedded identity data [41]. Generate a fresh event-specific link for each occasion; do not reuse one link for daily longitudinal sampling or assume a personal link is anonymous.

Validated task/sensor platform: collect raw audio, accelerometry or EEG under study consent. Store large streams in an approved encrypted repository. REDCap holds participant pseudonym, timestamps, device/firmware, checksum, location, QC flags, processing version and derived endpoint—not necessarily every raw sample.

Integration service: an institution-managed process maps an opaque survey-event token to study ID, checks duplicates and schema, imports only approved fields, logs transfers and reconciles failures. Keep direct identifiers and diagnoses out of URL parameters. Tokens remain sensitive because they can permit linkage; pseudonymized data are not automatically de-identified.

Biomarker layer: retain specimen collection and storage times, freeze–thaw history, assay platform/batch, PET tracer and processing version, and the interval between digital and biological measurements. A PET result obtained years before a digital task is not a contemporaneous reference.

Clinician workflow: research alerts should enter a defined review queue with coverage, response windows and escalation criteria. Tell participants whether the study is monitored for safety and whether exploratory findings will be returned. It must not silently imply emergency surveillance.

HIPAA and study governance

REDCap's official wording is that it **can be installed in environments supporting compliance** with HIPAA and other standards [39]. Qualtrics describes security controls and certifications [40]. Neither statement makes an arbitrary project or integration compliant.

Before collecting PHI, the institution must verify applicable contracts/BAAs, approved hosting and regions, encryption, identity/access management, least privilege, audit retention, messaging vendors, incident response, backups, retention/withdrawal policies and any AI processing. Voice can identify a person even after names are removed from transcripts. Consumer-device cloud services and analytics SDKs require their own review.

IRB approval, informed consent, capacity assessment where appropriate, caregiver roles and return-of-results policy are separate from technical security. HIPAA compliance is also distinct from validation for regulated trial data or diagnostic-device authorization.

11. A defensible study-development path

This is a proposed design derived from the evidence, not an existing standard protocol.

Stage A: verification and feasibility. Bench-test sensors; validate task timing; compare signals with video, gait laboratory, clinical EEG or other reference instruments. Pilot burden and accessibility before a long study.

Stage B: prospective biomarker cohort. Recruit a clinically relevant spectrum, including mimics and comorbidities. Establish baseline clinical, digital and disease-appropriate biological measures. Follow prespecified digital slopes and adjudicated outcomes; repeat biomarkers at clinically justified intervals. Calculate sample size from expected conversion/progression events, not from the number of sensor windows.

Stage C: locked external validation. Freeze hardware assumptions, preprocessing, features and model. Test new sites/devices and clinically relevant subgroups. Assess incremental value over existing tests and a low-cost baseline.

Stage D: utility trial. Compare usual care with digital-assisted triage or monitoring. Measure time to appropriate diagnosis, actionable referrals, missed disease, anxiety, workload, cost and patient outcomes. Registration should distinguish predictive accuracy from treatment benefit.

12. What is ready to build—and what is not

Most defensible near-term applications

1. **Clinical-trial prescreening:** use remote cognition or movement to prioritize confirmatory assessment, with explicit false-negative safeguards.

2. **Monitoring diagnosed disease:** quantify fluctuations, progression and treatment-associated functional change between visits.

3. **Biomarker-integrated triage:** combine an independently validated digital task with biological and clinical evidence, as illustrated by BioCog.

4. **Research infrastructure:** link raw signals, contextual EMA, clinical adjudication and assay provenance without collapsing them into one unexplained score.

A practical sequence is: quality-controlled baseline → persistent change → short active confirmation → clinician review → appropriate biomarker testing. The threshold and follow-up action must be validated for the particular condition; there is no single “decline score” that identifies its cause.

Unresolved questions

• Does a change signal predict pathology, symptoms, disability or merely activity opportunity?

• Does it generalize across language, education, disability, devices and healthcare settings?

• How many false alerts occur per patient-year?

• Does the signal add information beyond age, history, clinical testing and modern blood biomarkers?

• Can patients sustain the protocol without disproportionate caregiver work?

• Will earlier detection change available management enough to justify harms and costs?

Conclusion: the field has established meaningful digital measurements and several promising prognostic associations. It has not established a general consumer device that detects any neurological disease years before diagnosis. The strongest translation combines disease-specific sensing, repeated within-person assessment, biological confirmation and a tested clinical workflow.

13. Outcome-focused reference table

References 1–35 are scientific/clinical studies or reviews; 36–41 address research infrastructure. Trial-registry links are in Section 7. Review and protocol entries are labeled so they are not mistaken for primary outcome studies. Links use only permitted sites. Reported metrics are not directly comparable across populations or tasks.

Parkinson's: prediction and monitoring

Ref. / source	Design and finding	What it does not prove
1. Wearable movement-tracking data identify Parkinson's disease years before clinical diagnosis. Nature Medicine, 2023.	UK Biobank; 113 prodromal cases in reported comparison; up to seven-year lead time; prodromal AUPRC 0.07±0.03.	Prospective clinical screening utility or a high PPV at a chosen threshold.
2. Remote smartphone monitoring of Parkinson's disease and individual response to therapy. Nature Biotechnology, 2022.	mPower analysis; 960 participants with ≥5 active assessments; status AUC 0.80; motor-scale correlation r=0.71.	Confirmed diagnoses for all self-reported cases; general-population performance.
3. Predicting motor, cognitive & functional impairment in Parkinson's. Annals of Clinical and Translational Neurology, 2019.	237 early-PD participants; baseline smartphone features predicted 18-month clinical outcomes.	Independent multisite deployment performance.
4. WATCH-PD over 12 months. npj Parkinson's Disease, 2024.	82 early untreated PD and 50 controls; gait and tremor changed longitudinally.	Universal superiority to the total MDS-UPDRS; missingness constrained analysis.

Ref. / source	Design and finding	What it does not prove
5. Digital Outcomes as Biomarkers of Disease Progression in Early Parkinson's Disease. Movement Disorders, 2025. **Systematic review.**	15 longitudinal studies, 11 observational and four interventional; three technologies detected change when scales did not.	A pooled validated progression endpoint; methods remain heterogeneous.
6. Validation of Freezing-of-Gait Monitoring Using Smartphone. Telemedicine and e-Health, 2018.	32 PD patients; frequency-domain inertial CNN; sensitivity 93.8%, specificity 90.1%.	Prediagnostic PD detection or unrestricted free-living generalization.
7. Detecting Parkinson Disease Using a Web-Based Speech Task. JMIR, 2021.	726 participants; acoustic-feature/XGBoost model AUC 0.753 for self-reported PD.	Disease-specific diagnosis from speech alone.

Alzheimer's: cognition and pathological biomarkers

Ref. / source	Design and finding	Main limitation
8. Continuous Associations between Remote Self-Administered Cognitive Measures and Imaging Biomarkers. Journal of Prevention of Alzheimer's Disease, 2024.	684 adults; Mayo Test Drive associated with amyloid/tau PET and MRI measures.	Cross-sectional association, predominantly cognitively unimpaired cohort.

Ref. / source	Design and finding	Main limitation
9. Stricker Learning Span criterion validity. JINS, 2024.	353 adults; remote memory task comparable to in-person AVLT for PET-defined groups.	PET/testing timing mismatch; not autonomous diagnosis.

Ref. / source	Design and finding	Main limitation
10. Remote and in-clinic digital cognitive screening tools and cerebral amyloid status. Alzheimer's & Dementia: DADM, 2023.	69 cognitively normal adults; eight-day smartphone sampling; memory-task AUC 0.77.	Small sample; modest discrimination.
11. Monthly At-Home Computerized Cognitive Testing to Detect Diminished Practice Effects. Frontiers in Aging Neuroscience, 2021.	114 adults; reduced learning related to PET pathology; early slope predicted later cognitive decline, AUC 0.91.	Follow-up subset 72; requires independent replication.
12. Primary care detection of Alzheimer's disease using a self-administered digital cognitive test and blood biomarkers. Nature Medicine, 2025.	BioCog developed in 223; independently evaluated in 403 across 19 primary-care centers; combined AD accuracy 90%.	Clinical proof-of-concept, not asymptomatic population screening.
13. Prediction of Alzheimer's disease progression within 6 years using speech. Alzheimer's & Dementia, 2024.	166 MCI participants; language-model features plus demographics; accuracy 78.5%, sensitivity 81.1%.	Already-impaired cohort; interview transcripts rather than everyday passive listening.
14. Smartwatch- and smartphone-based remote assessment of brain health and detection of MCI. Nature Medicine, 2025.	Intuition; publication reports 23,004 adults; initial interactive-cognition MCI classification.	Does not establish passive-watch AD prediction; registry enrollment differs.

Ref. / source	Design and finding	Main limitation
15. RADAR-AD: assessment of multiple remote monitoring technologies. Alzheimer's Research & Therapy, 2025.	Six remote technologies; nested cross-validation; prodromal-stage AUC 0.73.	Stage discrimination is not future conversion prediction or clinical benefit.
16. Sleep as a Potential Biomarker of Tau and β-Amyloid Burden. Journal of Neuroscience, 2019.	Sleep EEG and PET; tau associated with impaired oscillation–spindle coupling, amyloid with diminished slow-wave activity.	Polysomnography, not consumer-headband validation.
17. Coupled sleep rhythm disruption predicts cognitive decline in Alzheimer's disease. Science Bulletin, 2025.	93 adults; EEG, CSF, MRI and two-year cognition; disrupted coupling associated with subsequent decline.	Selected research cohort; no validated general-population alarm threshold.
18. A self-driven approach for multi-class discrimination in AD based on wearable EEG. Computer Methods and Programs in Biomedicine, 2022.	Commercial 16-channel EEG; automated artifacts/features; MLP; leave-one-subject-out F1=0.88.	Preliminary cross-sectional classification, not years-ahead prognosis.

MS and clinical interpretation

Ref. / source	Design and finding	Main limitation
19. A smartphone sensor-based digital outcome assessment of MS. Multiple Sclerosis, 2022.	Floodlight; ICC 0.61–0.85; cognition correlation r=0.82; clinical/MRI associations.	Functional validation, not proof of preclinical MS diagnosis.

Ref. / source	Design and finding	Main limitation
20. Biosensor vital sign detects MS progression. Annals of Clinical and Translational Neurology, 2021.	68 patients; IMU/EMG tapping metrics; relapsing versus progressive AUROC 0.93/0.96.	Small sample; individual progression decisions need validation.
21. Long-term unsupervised mobility assessment in movement disorders. The Lancet Neurology, 2020. **Review.**	Explains disparities between supervised and everyday mobility measurement.	Not a new diagnostic-accuracy trial; contextual evidence.
22. Design of AT-HOME PD. Annals of Clinical and Translational Neurology, 2021. **Design/feasibility.**	226 trial alumni enrolled; 80% downloaded app, 71% completed online patient reports.	Enrollment feasibility, not completed progression-validation results.
23. Wearable sensors for early diagnosis of Parkinson disease. JMIR, 2025. **Systematic review.**	12 eligible studies; ten cross-sectional; IMUs/accelerometers predominated.	Limited longitudinal and standardized diagnostic evidence.
24. Home-based digital biomarkers for MCI to mild AD. JMIR, 2019. **Systematic review.**	26 eligible studies; small, heterogeneous real-world evidence base.	Historical synthesis, not a current approved-device inventory.

Validation frameworks and broader neurological applications

Ref. / source	Design and finding	Main limitation
25. V3: verification, analytical validation and clinical validation. npj Digital Medicine, 2020. **Framework.**	Defines fit-for-purpose evidence for biometric monitoring technologies.	Framework is not validation of a particular device.
26. V3+ usability validation. npj Digital Medicine, 2025. **Framework.**	Adds user-centricity and scalability to V3.	No disease-specific diagnostic accuracy result.
27. Remote digital monitoring in Huntington disease. JMIR, 2022.	219 participants across groups; ICC 0.89–0.98; cognitive correlations 0.79–0.90.	Short-period reliability/cross-sectional validity, not drug efficacy.
28. PROSA speech biomarkers in ALS/FTD. ALS and Frontotemporal Degeneration, 2023. **Protocol.**	Plans low-burden telephone speech assessment against motor, respiratory and cognitive measures.	Planned outcomes; no accuracy claim from this protocol.
29. Wearable motion capture predicts progression in Friedreich's ataxia. Nature Medicine, 2023.	Nine patients/nine controls; motion features improved nine-month prediction of selected clinical scores.	Very small sample; research-grade suit.
30. Full-body movement predicts DMD trajectory. Nature Medicine, 2023.	21 patients/17 controls; 17-sensor suit; 12-month behavioral biomarker development.	Progression research, not consumer screening.

Interventions, raw-signal methods and usability

Ref. / source	Design and finding	Main limitation
31. Gamma sensory stimulation in mild probable AD. PLOS ONE, 2022.	EEG entrainment feasibility and 15-person randomized pilot; exploratory MRI, memory and activity-rhythm findings.	Not definitive efficacy or human amyloid/tau-lowering evidence.
32. Soft robotic apparel to avert freezing of gait. Nature Medicine, 2024.	One-person repeated trial; indoor freezing 0% with assistance versus 39±16% unassisted.	n-of-1 proof of concept; not early diagnosis.
33. Self-supervised wrist accelerometry for gait detection in older adults. Scientific Reports, 2024.	ElderNet; >1,000 unlabeled and 83 labeled participants; accuracy 96.43%, F1 82.92%.	Gait-event classification is not disease classification.
34. Patient experience and feasibility of remote monitoring in Parkinson's. Movement Disorders Clinical Practice, 2024.	200 patients; 58% one-year completion; median 17.5 wear-hours/day; technical dropout prominent.	Device/study-specific; not universal adherence.
35. Floodlight Open user experience. JMIR Human Factors, 2024.	Simplified activation: 53.9% to 74.6%; reminders associated with improved engagement.	UX comparisons do not validate a diagnostic algorithm.

Research infrastructure—not diagnostic-outcome evidence

Ref. / source	Supported finding or capability	Qualification
36. REDCap methodology and workflow. Journal of Biomedical Informatics, 2009.	Metadata-driven electronic research data capture.	Foundational paper outside the ten-year clinical window.
37. The REDCap consortium. Journal of Biomedical Informatics, 2019.	Describes multicenter platform dissemination and governance.	Not proof that a specific deployment is compliant.
38. Qualtrics-based mobile research without an app. Frontiers in Digital Health, 2022. **Methods.**	Describes surveys, messaging and tasks using a hypothetical smoking study.	Not a neurological Qualtrics–REDCap validation trial.
39. REDCap official software documentation.	Surveys, databases, audit trails, exports and deployment options supporting regulatory compliance.	Institutional hosting/configuration determines compliance.
40. Qualtrics security documentation.	Describes security controls, encryption and certifications.	Verify actual licensed environment, contracts and PHI authorization.
41. Qualtrics personal links.	Recipient-linked, unique, single-use survey links; license-dependent functionality.	Not anonymous by default; repeated EMA needs event-level link management.

14. Reading the metrics

• **ROC AUC:** discrimination across sensitivity/specificity thresholds; does not establish calibration or PPV.

• **AUPRC:** precision–recall area; strongly affected by class prevalence. Do not compare directly with ROC AUC.

• **Sensitivity / specificity:** true-positive / true-negative rates at a stated threshold.

• **PPV:** probability that a positive result is truly positive in the tested population.

• **ICC:** reliability across repeat measurements; high reliability does not guarantee disease validity.

• **Correlation:** association; not interchangeability, agreement or causation.

• **F1:** harmonic mean of precision and recall; the class definition and averaging method matter.

• **PET / CSF:** positron-emission tomography / cerebrospinal fluid. Tracers and analytes measure different biological processes.

• **EMA:** ecological momentary assessment; often self-report, not necessarily a cognitive performance test.

15. Review limitations

The report is strongest for Parkinson’s, AD/MCI and MS; coverage of ALS, ataxias and other causes of bodily decline is selective. It is not an exhaustive search of all registered trials, all publishers or all countries. Keyword retrieval can miss substudies and can return irrelevant trials. No formal risk-of-bias scoring or meta-analysis was performed.

Many papers were assessed from indexed abstracts rather than full methods and supplements. Enrollment, follow-up and registry/publication discrepancies are reported where identified, not resolved by assumption. Some influential evidence uses overlapping cohorts. Older benchmark models are not rankings of current commercial devices.

Current trial labels and documentation were checked on 20 October 2026. Registry updates may lag actual conduct, and no-results-posted is distinct from no published evidence. The report does not assess regulatory authorization or recommend personal screening with a particular product.

Most important next step for a development team: select one disease, one intended population, one actionable decision and one clinically meaningful endpoint before selecting the device or model.